



# COFFY COIN

MODULAR BLOCKCHAIN ECOSYSTEM

WHITEPAPER | 2026

*"Brewing the Future of Coffee with Blockchain"*

Drink-to-Earn | Play-to-Earn | SocialFi | Base Mainnet (ERC-20)

**15B**

Max Supply

**35%**

Community

**50%**

Max APY

**5**

Characters

**7+**

Games

CoffyCore | Base Mainnet

**0x29248bA2420757bF50595Af6d8903E5d8Dcb9b41**

Verified Contract

Base Mainnet

OpenZeppelin

Anti-Sybil

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01

## Executive Summary

**15B**  
Max Supply

**Base**  
Network

**50%**  
Max APY

**7+**  
Live Games

**5**  
Characters

Coffy Coin (COFFY) is a next-generation Play-to-Earn and SocialFi token on Base Mainnet. The modular ecosystem combines an immutable core contract with plug-and-play modules for gaming, staking, activities, and future DAO governance. COFFY is the first Drink-to-Earn, Play-to-Earn, and SocialFi coin in the coffee-gaming vertical.

The architecture is built for longevity: the core is immutable after deploy — protecting all token holders from rug-pull risk — while modules upgrade independently. Multi-signature control, timelocked parameter changes, and ECDSA-signed reward claims form a layered security model exceeding GameFi industry standards.

| Highlight            | Description                                                                                     |
|----------------------|-------------------------------------------------------------------------------------------------|
| Immutable Core       | Total supply, allocations, and role logic permanently locked on-chain — no upgradeable proxy.   |
| Sustainable Emission | 2.5% semi-annual inflation (5% annual max) — far below competitors, preserving long-term value. |
| Dynamic APY          | Staking scales with total staked and character multipliers — up to 50% APY, zero minimum stake. |
| 7 Live Games         | Fully operational gaming ecosystem generating real engagement and on-chain activity today.      |
| Cross-Chain Ready    | Bridge role and daily mint limits built into core — BNB, Polygon, Arbitrum launching Q1 2026.   |

## Project Overview

Coffy Coin bridges the global coffee culture — a \$500B+ industry — with blockchain, creating an ecosystem where users earn through gameplay, staking, social participation, and real-world coffee engagement. The mission: make DeFi and Web3 accessible to everyday people through the universally loved ritual of coffee.

| Pillar        | Description                                                                     |
|---------------|---------------------------------------------------------------------------------|
| Drink-to-Earn | Real-world coffee photo verification rewards via on-chain Snap system           |
| Play-to-Earn  | 7+ games with on-chain stakes, PvP, and weekly reward caps up to 50K COFFY      |
| Stake-to-Earn | Dynamic APY 2% to 50% with 7-day lock and character multipliers applied         |
| Move-to-Earn  | Step reward system — oracle-verified, daily and weekly limits enforced on-chain |
| SocialFi      | Referral system, DAO governance (Q2 2026), creator economy, global leaderboards |
| Cross-Chain   | Bridge infrastructure for BNB Chain, Polygon, and Arbitrum — launching Q1 2026  |

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Smart Contract Architecture

The modular design separates an immutable core from upgradeable modules. The core holds all token logic, role management, and treasury functions. Modules interact through a strict ICoffyCore interface — ensuring the core can never be manipulated by module-side bugs. Each module is independently pausable and subject to per-module weekly spending caps.

| Contract                | Role            | Description                                                |
|-------------------------|-----------------|------------------------------------------------------------|
| CoffyCoin Core          | Immutable Token | Module registry, bridge, inflation, per-module spend caps  |
| Game Module             | Play-to-Earn    | PvP/Battle, single games, quick match, characters, streaks |
| Activity Module         | Staking & Earn  | Dynamic APY, session/step/snap rewards, referral, XP       |
| Timelock Module         | Governance      | Time-delayed parameter changes                             |
| Bonus Module            | Rewards Hook    | onGameWin / onStakingReward callback hooks                 |
| DAO Module (Q2)         | Governance*     | Legendary Dragon holders vote on protocol                  |
| NFT Module (Q2)         | Characters*     | On-chain character NFTs with rarity tiers                  |
| Cross-Chain Bridge (Q1) | Interop*        | BNB, Polygon, Arbitrum — 1%/day mint cap                   |

|                 |                                            |
|-----------------|--------------------------------------------|
| CoffyCoin Core  | 0x29248bA2420757bF50595Af6d8903E5d8Dcb9b41 |
| Game Module     | 0xEb00A304DD1aB9A5bC995d4eD9cAFc190bC593Ea |
| Activity Module | 0x1084Ba72eaF89E4Ed0c0320FDB4C6A51159c15eb |

Deployed contracts — Base Mainnet (Verified on BaseScan)

\* Planned — not yet deployed

Role-Based Access Control

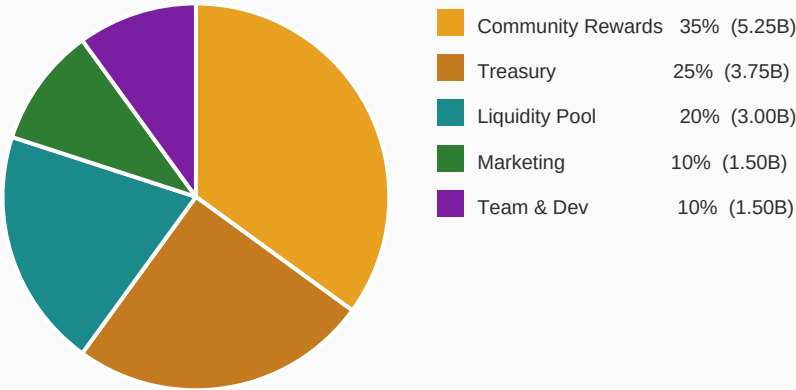
| Role          | Holder              | Permissions                                                  |
|---------------|---------------------|--------------------------------------------------------------|
| DEFAULT_ADMIN | Multisig            | Full admin — permanently revoked from deployer after 30 days |
| ADMIN_ROLE    | Multisig / Timelock | Module registration, parameter configuration                 |
| MODULE_ROLE   | Registered Modules  | payReward, burnFrom, mintFor — weekly caps enforced          |
| TIMELOCK_ROLE | TimelockModule      | Delayed critical parameter updates                           |
| KEEPER_ROLE   | Oracle / Bot        | Inflation trigger, maintenance tasks                         |
| BRIDGE_ROLE   | Bridge Contract     | burnForBridge / mintFromBridge — 1% per day cap              |
| ORACLE_ROLE   | Backend Signer      | ECDSA claim signature verification                           |

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Token Economics

| Parameter          | Value                                                      |
|--------------------|------------------------------------------------------------|
| Token Name         | Coffy Coin                                                 |
| Symbol             | COFFY                                                      |
| Network            | Base Mainnet (Coinbase L2, ERC-20)                         |
| Max Supply         | 15,000,000,000 (15B) COFFY                                 |
| Decimals           | 18                                                         |
| CoffyCore Contract | 0x29248bA2420757bF50595Af6d8903E5d8Dcb9b41                 |
| Game Module        | 0xEb00A304DD1aB9A5bC995d4eD9cAFc190bC593Ea                 |
| Activity Module    | 0x1084Ba72eaF89E4Ed0c0320FDB4C6A51159c15eb                 |
| Inflation          | 2.5% semi-annual (5% annual max) via triggerInflation()    |
| Weekly Spend Cap   | 1% of current pool balance — dynamic, self-balancing       |
| Bridge Mint Limit  | 1% of total supply per day — anti-infinite-mint protection |

Token Distribution



| Allocation        | Tokens        | Share | Purpose                                 |
|-------------------|---------------|-------|-----------------------------------------|
| Community Rewards | 5,250,000,000 | 35%   | Game rewards, staking, ecosystem growth |
| Treasury          | 3,750,000,000 | 25%   | Operations, development, partnerships   |
| Liquidity Pool    | 3,000,000,000 | 20%   | DEX liquidity and price stability       |
| Marketing         | 1,500,000,000 | 10%   | User acquisition and brand campaigns    |
| Team & Dev        | 1,500,000,000 | 10%   | Core team — subject to vesting module   |

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## Game Ecosystem & Play-to-Earn

The Game Module handles three paradigms: Solo Games (vs. oracle), PvP Matches (on-chain stakes between two players), and Battles (asynchronous challenge mode). All reward claims are ECDSA-signed by a trusted backend oracle, preventing on-chain manipulation. A 2-minute minimum session is enforced via block timestamps.

| Game               | Mode               | Max Weekly        | Description                               |
|--------------------|--------------------|-------------------|-------------------------------------------|
| Coffee Chess       | PvP Multiplayer    | 50,000 COFFY/week | Classic chess with on-chain stakes        |
| Coffee Checkers    | PvP Multiplayer    | 50,000 COFFY/week | Room ID or Quick Match system             |
| Bee Game Adventure | 3D Adventure       | 50,000 COFFY/week | Open world, NPC combat, flight            |
| FlagRacer Online   | Multiplayer Racing | 50,000 COFFY/week | Dynamic tracks, tournaments               |
| Coffy in Maze      | 3D Puzzle          | 50,000 COFFY/week | Procedural layouts, increasing difficulty |
| Coffyverse City3D  | Strategy Co-op     | 50,000 COFFY/week | Tactical missions, upgrades               |
| Coffy Adventure    | Action RPG         | 50,000 COFFY/week | Bosses, power-ups, leaderboard            |

### Reward Parameters

| Parameter          | Value         | Notes                                   |
|--------------------|---------------|-----------------------------------------|
| Min Session        | 2 minutes     | Enforced on-chain via block timestamp   |
| Base Weekly Cap    | 50,000 COFFY  | Scales 2x-10x with character multiplier |
| Daily User Limit   | 10,000 COFFY  | Default cap — configurable by admin     |
| Max Daily Reward   | 500,000 COFFY | Anti-whale cap per user per day         |
| Max Daily Bets     | 20 bets       | Anti-grinding protection per wallet     |
| Match Protocol Fee | 5%            | Taken from PvP stakes — configurable    |
| Quick Match Expiry | 10 minutes    | Stake returned if no match found        |

## Staking & DeFi Mechanics

The Activity Module implements a dynamic APY formula — APY scales inversely with total staked tokens, rewarding early stakers while preventing hyperinflation as adoption grows. Five additional earning mechanisms complement staking for a comprehensive DeFi experience.

| Parameter            | Value          | Notes                                                      |
|----------------------|----------------|------------------------------------------------------------|
| Minimum APY          | 2%             | Floor rate — guaranteed regardless of pool size            |
| Maximum APY          | 50%            | Achieved at low total staked and high character multiplier |
| Lock Period          | 7 Days         | Rewards only claimable after full lock period              |
| Early Exit Penalty   | 5%             | Burned — reduces circulating supply permanently            |
| Minimum Stake        | None           | Open to all — zero barrier to entry                        |
| Character Multiplier | 2x to 10x      | Applied directly to APY calculation                        |
| Reward Source        | Community Pool | Weekly cap: 1% of current pool balance                     |

### Earning Mechanisms

| Mechanism       | Description                                                                |
|-----------------|----------------------------------------------------------------------------|
| Session Rewards | Standard play session claims — oracle signed, daily/weekly limits enforced |
| Step Rewards    | Mobile step tracking — minimum step count verified in oracle signature     |
| Snap Rewards    | Coffee photo verification — per-user cooldown enforced on-chain            |
| Referral System | On-chain tracking — rewards to both referrer and referee                   |
| XP & Badges     | Persistent XP and badge system stored in core contract                     |

## Security & Anti-Sybil Framework

Layered defense covers smart contract security, sybil resistance, economic attack prevention, and operational security through multisig and timelock controls.

| Protection             | Description                                                                                                                |
|------------------------|----------------------------------------------------------------------------------------------------------------------------|
| ECDSA Signed Claims    | All reward claims require a valid oracle signature. Each signature is single-use (replay protection via usedSigs mapping). |
| Wallet Age Check       | Minimum 3-day wallet age prevents fresh-wallet sybil attacks on all claim functions.                                       |
| Min Balance Option     | Optional COFFY minimum balance before claiming — incentivizes long-term holding.                                           |
| Per-Module Caps        | Independent weekly spend cap per module (15M game / 1M activity recommended). A compromised module cannot drain the pool.  |
| Dynamic Pool Limits    | 1% weekly spend cap self-adjusts with pool size — prevents runaway drain automatically.                                    |
| Immutable Core         | No upgradeable proxy — supply, allocations, and roles cannot change after deploy.                                          |
| 30-Day Deployer Expiry | Deployer admin rights permanently expire 30 days post-launch via immutable deployerExpiry value.                           |
| Emergency Pause        | All contracts extend OpenZeppelin Pausable. Only multisig can unpause — admin alone cannot.                                |
| Reentrancy Guards      | All reward functions use nonReentrant modifier — blocks reentrancy-based drain attacks.                                    |
| Recovery Address       | Immutable recovery address with cancelRecovery() safety valve for accidental triggers.                                     |

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## Character System & Multipliers

Characters are purchased with COFFY (creating buy pressure and deflationary burn mechanics) and grant reward multipliers across all earning activities. The full system is stored on-chain via the userCharacters mapping and getCharacterMultiplier() function.

| Tier                      | Multiplier | Cost       | Benefits                                         |
|---------------------------|------------|------------|--------------------------------------------------|
| Tier 1 — Common           | 2x         | Low        | Base multiplier — accessible to all users        |
| Tier 2 — Uncommon         | 3x - 4x    | Medium-Low | Enhanced game rewards and staking APY            |
| Tier 3 — Rare             | 5x - 6x    | Medium     | Priority matchmaking, bonus pool access          |
| Tier 4 — Epic             | 7x - 8x    | High       | Tournament eligibility, premium reward pools     |
| Tier 5 — Legendary Dragon | 10x        | Very High  | DAO membership for protocol governance (Q2 2026) |

### Economic Loop

Character purchase burns COFFY or transfers to community pool. Virtuous cycle: higher character demand -> more COFFY burned -> reduced supply -> price appreciation -> new users join and buy characters. Legendary Dragon holders gain DAO governance rights in Q2 2026.

## Cross-Chain & Bridge Infrastructure

The core contract ships with a production-ready bridge interface. `BRIDGE_ROLE` is a dedicated access control role allowing certified bridge contracts to mint and burn COFFY in a controlled, rate-limited manner. A 1%/day mint cap prevents infinite mint attacks even in a worst-case bridge exploit scenario.

| Function                          | Direction        | Limit             | Notes                                                      |
|-----------------------------------|------------------|-------------------|------------------------------------------------------------|
| <code>burnForBridge()</code>      | Base to External | No limit          | Burns COFFY on Base when user bridges out                  |
| <code>mintFromBridge()</code>     | External to Base | 1% supply/day     | Mints COFFY when user arrives from another chain           |
| <code>setBridgeContract()</code>  | Admin            | Timelock required | Auto-revokes old bridge <code>BRIDGE_ROLE</code> on update |
| <code>dailyBridgeMintLimit</code> | Config           | 1% total supply   | Hard cap — prevents infinite mint on bridge exploit        |

### Target Chains — Q1 2026

| Chain        | Type          | Rationale                                       |
|--------------|---------------|-------------------------------------------------|
| BNB Chain    | Lock-and-Mint | Largest GameFi user base in Asia-Pacific        |
| Polygon PoS  | Lock-and-Mint | Low fees, established DeFi and gaming ecosystem |
| Arbitrum One | Lock-and-Mint | Ethereum-aligned L2, growing DeFi TVL           |

# 2026 Roadmap

|              |                                                                                                                            |
|--------------|----------------------------------------------------------------------------------------------------------------------------|
| Q1 — Active  | AI Game Matchmaking & Anti-Cheat   Cross-Chain Bridge (BNB, Polygon, Arbitrum)   Character NFT System with Rarible         |
| Q2 — Planned | Coffyverse Metaverse — Virtual Coffee Shops   Social Tournaments (1M COFFY prizes)   Creator Economy   DAO Governance      |
| Q3 — Future  | Yield Farming & Liquidity Mining   COFFY Launchpad for Gaming Projects   Enterprise API for Coffee Shops   NFT Marketplace |
| Q4 — Vision  | Global Championships (\$500K+ Prize Pool)   AR/VR Coffee Gaming   COFFY Debit Card & Payments   Major Gaming Partnerships  |

Current Deployment — Base Mainnet (All Contracts Verified)

CoffyCoin Core: 0x29248bA2420757bF50595Af6d8903E5d8Dcb9b41

Game Module: 0xEb00A304DD1aB9A5bC995d4eD9cAFc190bC593Ea

Activity Module: 0x1084Ba72eaF89E4Ed0c0320FDB4C6A51159c15eb

All 7 games live. Staking active with up to 50% APY. Join: [t.me/+GBhxiB8egWY1M2Q8](https://t.me/+GBhxiB8egWY1M2Q8)

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Team & Governance

COFFY is governed by a multisig wallet holding ADMIN\_ROLE. Critical parameter changes route through Timelock Module enforcing time-delayed execution. The deployer's admin rights automatically and permanently expire 30 days after launch — an on-chain guarantee.

| Layer            | Timeline | Description                                                       |
|------------------|----------|-------------------------------------------------------------------|
| Multisig         | Current  | Holds ADMIN_ROLE — all critical changes require multisig approval |
| Timelock         | Current  | Delays sensitive parameter changes — prevents rushed decisions    |
| DAO Governance   | Q2 2026  | Community treasury management — Legendary Dragon holder voting    |
| Quadratic Voting | Future   | Evolved voting to balance large vs small token holders            |

DAO Details (Q2 2026)

- Legendary Dragon holders receive automatic DAO membership via on-chain isDAOMember mapping
- DAO module manages community treasury spend proposals — voting weight proportional to holdings
- Protocol parameters (APY bounds, game fees, spend caps) require DAO vote to change
- Initial: 1-token-1-vote governance, evolving to quadratic voting in later phases

Ecosystem Partners

Strategic partnerships with coffee brands (RoastBrew, Javamingle, Cafenest, PerkCafe) for real-world integrations, alongside Web3 infrastructure partners including Unity, BNB Chain, Polygon, OpenSea, Animoca Brands, and Base for distribution and tooling.

| Channel     | Link                  | Purpose                                                          |
|-------------|-----------------------|------------------------------------------------------------------|
| Website     | www.coffycoin.xyz     | Main hub — games, staking, tokenomics                            |
| Telegram    | t.me/+GBhxb8egWY1M2Q8 | Community chat — announcements and support                       |
| Twitter / X | x.com/coffycoinxyz    | News, updates, and ecosystem developments                        |
| BaseScan    | basescan.org          | Core: 0x2924...b41   Game: 0xEb00...3Ea   Activity: 0x1084...5eb |

|               |                       |                    |
|---------------|-----------------------|--------------------|
| coffycoin.xyz | t.me/+GBhxb8egWY1M2Q8 | x.com/coffycoinxyz |
|---------------|-----------------------|--------------------|

*This whitepaper is for informational purposes only and does not constitute financial or investment advice. COFFY tokens are utility tokens for use within the Coffy Coin ecosystem. Always conduct your own due diligence.*